

CHAPTER 8

Web Services

WWW · web page and website · URL · domain names · browser and web server · search engines · web hosting · HTML and CSS · XML · web services

QUICK TRICK

Learn the fast tricks first (details below):

1. **Internet** = the network of networks. **WWW** = the web pages on it (invented by **Tim Berners-Lee**). Web needs: **browser + server + HTTP + HTML + URL**.
2. **URL** = scheme://host:port/path?query#fragment. Default ports: http 80, https 443.
3. **HTML** displays data with **predefined** tags. **XML** describes data with **user-defined** tags and strict rules.
4. **Static page** = same for everyone. **Dynamic page** = made by a server script and database for each request.

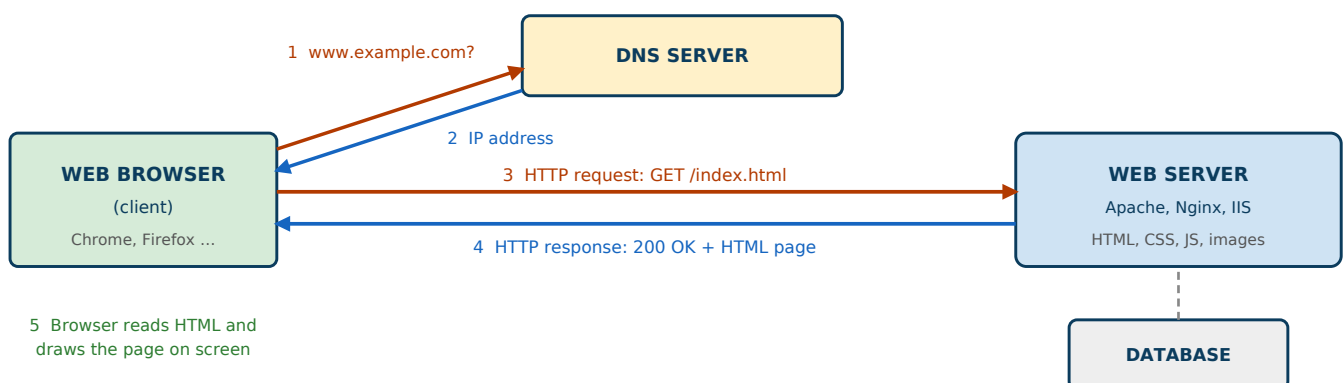
8.1 World Wide Web: web page, website, web portal

● LEVEL 1 · BEGINNER

The problem. The Internet (Chapter 1) joins computers, but a normal person needs an easy way to find and read information. **The idea.** The **World Wide Web (WWW)**, built by **Tim Berners-Lee** at **CERN** (1989–91), is a system of linked documents called **web pages**. Clicking a **hyperlink** in one page takes you to another page, on any server in the world. The Web is one **service** that runs on the Internet (e-mail and FTP are others).

Term	Meaning
Hypertext	Text that contains links to other text. Hypermedia also links pictures, audio and video
Web page	A single document written in HTML and shown in a browser
Website	A collection of related web pages under one domain name (for example all pages of a school)
Home page	The first (main) page of a website, usually index.html
Web portal	A website that gathers many services in one place (news, mail, search, login), for example a government service portal
Hyperlink	Clickable text or image that opens another page or resource

What is needed for the Web? A **web browser** (client), a **web server**, the **HTTP** protocol (Chapter 6), **HTML** to write pages, and **URLs** to name them. The picture shows what happens when you open a page.



Static and dynamic web pages

Point	Static web page	Dynamic web page
Content	Fixed; same for every visitor until the file is edited	Created when requested; can differ by user, time, search
Made with	HTML, CSS (and some JavaScript)	Server-side scripts (PHP, ASP.NET, JSP, Python, Node.js) with a database
Speed and cost	Fast, cheap to host	Slower, needs a more powerful server
Update	Edit the HTML file	Change data in the database; page updates itself
Examples	School brochure, personal portfolio	Online shop, social media, railway booking, result pages

Client-side script (JavaScript, created 1995) runs **inside the browser** (menus, form checks, animation).

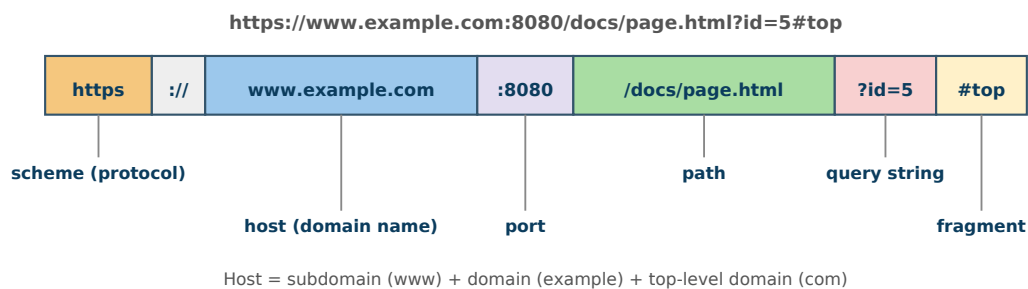
Server-side script runs **on the web server** and sends only the finished HTML to the browser.

Web version	Idea
Web 1.0	Read-only, static pages; a few authors, many readers
Web 2.0	Read-write; user-created content, blogs, social media, wikis, video sharing
Web 3.0	Semantic and decentralised web; machines understand meaning of data, more user control

8.2 URL: Uniform Resource Locator

● LEVEL 1 · BEGINNER

A **URL** is the **address of a resource** (page, image, file) on the Web. Typing it in the address bar tells the browser which server to contact, with which protocol, and what to ask for.



Part	In the example	Meaning
Scheme (protocol)	https	Which protocol to use: http, https, ftp, mailto, file
Host (domain name)	www.example.com	Server name: subdomain www + domain example + TLD com . An IP address can be used instead
Port	:8080	Optional. If missing, the default is used: http 80, https 443, ftp 21
Path	/docs/page.html	Folder and file on the server
Query string	?id=5	Data sent to the server (name=value pairs, joined by &)
Fragment	#top	A position inside the page; handled by the browser, not sent to the server

Absolute URL	Relative URL
Full address: https://www.example.com/images/logo.png	Only the path from the current page: images/logo.png

URI, URL and URN. A **URI** (Uniform Resource Identifier) is the general name for any identifier of a resource. A **URL** says **where** it is (location) and how to get it. A **URN** names a resource by a permanent name only. Every URL is a URI; not every URI is a URL.

COMMON MISTAKE

The **domain name** (example.com) is only **part** of a URL. A URL also has the protocol, path and more. Also, **http://** and **https://** are different: https is the encrypted version (port 443).

8.3 Domain names and registration

● LEVEL 1 · BEGINNER

Humans use names, computers use IP addresses, and **DNS** (Chapter 6) translates between them. A **domain name** is the unique, human-readable name of a website, read from right to left: **www.example.com** = host/subdomain **www**, domain **example**, top-level domain **.com**.

Kind	Examples	Note
Generic TLD (gTLD)	.com, .org, .net, .info, .biz	Open to most users
Sponsored TLD	.edu, .gov, .mil, .int	For specific groups (education, government, military)
Country-code TLD (ccTLD)	.in, .uk, .us, .jp	Two letters, one per country. .in is managed by NIXI
India second level	.co.in, .org.in, .net.in, .ac.in, .edu.in, .gov.in, .nic.in, .res.in	Used by business, organisations, academic and government bodies

Domain name rule	Detail
Allowed characters	Letters (a-z), digits (0-9) and hyphen (-). No spaces, no underscore
Hyphen	Cannot be at the start or end of a part
Case	Not case-sensitive (Example.COM = example.com)
Length	Each part (label) up to 63 characters; the whole name up to 253 characters

How to get a domain. (1) Check that the name is free. (2) Choose a **domain registrar** (a company accredited by **ICANN** or by the country registry) and pay a **yearly fee**. (3) Give your **name servers** (NS) or point the A record to the IP of your web host. The domain is **rented**, not bought forever, and must be renewed. **WHOIS** is a public record of who registered a domain.

BPSC POINT

ICANN (set up in 1998) coordinates domain names and IP addresses at the top level. **Domain name ≠ website ≠ web hosting**: the domain is the address, the hosting is the land where the files live, the website is the content.

8.4 Web browser and web server

● LEVEL 1 · BEGINNER

A **web browser** is the client program that **requests, receives and displays** web pages. It sends HTTP requests, reads the HTML, CSS and JavaScript it receives, and **renders** (draws) the page. Examples: **Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Opera**.

Browser feature	Meaning
Address bar	Where you type a URL or a search
Tabs	Open several pages in one window
Bookmarks / favourites	Saved links to pages you like
History	List of pages visited
Cache	Copies of pages and images saved on your device so that repeat visits are faster
Cookies	Small text files a website stores in the browser to remember you (login, cart, preferences)
Incognito / private mode	Does not save history, cookies or form data on your device after you close it (websites and the ISP can still see you)
Extensions / add-ons	Small programs that add features (ad blocker, password manager)

Cookies: **session** cookies vanish when the browser is closed; **persistent** cookies stay until they expire. Cookies make the stateless HTTP protocol remember users (Chapter 6).

A **web server** is a computer plus **software** that stores websites and answers browser requests. The software listens on **port 80 (HTTP) and 443 (HTTPS)**, finds the file or runs the script asked for, and sends back the response. Popular web server software: **Apache HTTP Server, Nginx, Microsoft IIS** (Internet Information Services), LiteSpeed.

Point	Web browser	Web server
Role	Client; asks for pages	Server; gives pages
Runs on	User's device	A server computer, always on
Examples	Chrome, Firefox, Edge	Apache, Nginx, IIS
Speaks	Sends HTTP requests, shows HTML	Receives requests, sends HTTP responses

8.5 Search engines

● LEVEL 1 • BEGINNER

A **search engine** is a website that finds pages for a word or phrase you type. Examples: **Google, Bing, DuckDuckGo, Yahoo**. It is **not** the same as a browser: the browser opens the search engine's website.

Component	Job
Crawler (spider, bot)	Visits web pages automatically, follows links, and collects their content
Indexer	Stores the words found on each page in a huge index (like the index at the back of a book)
Query processor and ranking	Looks up your words in the index and sorts results by relevance. Google's early method was PageRank (Larry Page and Sergey Brin): pages linked from many good pages rank higher

Search tip	Effect
"computer network" (in quotes)	Exact phrase
network -wifi	Exclude a word
site:gov.in census	Search only inside one site
filetype:pdf bihar	Find only PDF files

SEO (Search Engine Optimization) means improving a website so that it appears higher in search results. A **web directory** is a list of sites grouped by topic, made by people.

8.6 Web hosting

● LEVEL 1 • BEGINNER

Web hosting is a service that gives you space on a **web server** that is always connected to the Internet, so that your website can be seen by everyone. A company that does this is a **hosting provider**. You upload your files (usually with **FTP / SFTP** or a control panel such as cPanel) and connect your **domain name** to the server.

Hosting type	Meaning	Cost / control	Best for
Shared	Many websites share one server and its resources	Cheapest; least control; one busy site can slow others	Small sites, blogs
VPS (Virtual Private Server)	One physical server split into virtual servers, each with its own share	Medium	Growing business sites
Dedicated	The whole server is rented for one customer	Costly; full control	Large, heavy-traffic sites
Cloud	Site runs on many connected servers; resources grow or shrink as needed, pay for use	Flexible	Sites with changing traffic
Colocation	You own the server but keep it in the provider's data centre	You pay for space, power, bandwidth	Organisations with their own hardware

Things to compare: storage, bandwidth (data transfer), **uptime** (for example 99.9% means about 8.8 hours of downtime in a year), security, backup and support.

Step	Put a website online
1	Write the website (HTML, CSS, scripts, database)
2	Register a domain name
3	Buy a hosting plan
4	Upload the files to the server (FTP / SFTP)
5	Point the domain to the host (name servers or A record). The website is now live

8.7 HTML: HyperText Markup Language

● LEVEL 2 · EXAM

HTML is the standard **markup language** for writing web pages. It is **not a programming language**: it does not calculate or decide; it only **describes the structure** of a page. Browsers read the **tags** and decide how to show the content. The current version is **HTML5** (W3C recommendation in 2014). File extension: **.html** or **.htm**.

Term	Meaning	Example
Tag	A keyword in angle brackets	<p>
Element	Start tag + content + end tag	Hello
Attribute	Extra information inside the start tag, as name="value"	
Container (paired) tag	Has a start tag and an end tag	 ...
Empty (void) tag	No content, no end tag	 , <hr>, , <input>, <meta>, <link>
Comment	Ignored by the browser	<!-- note -->

Structure of an HTML document

HTML code

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Page</title>
</head>
<body>
  <h1>Welcome</h1>
  <p>This is a <b>simple</b> page.</p>
  <ul>
    <li>Computer Networks</li>
    <li>Digital Logic</li>
  </ul>
  <a href="https://example.com">Visit site</a>
  
</body>
</html>
```

Part	Job
<!DOCTYPE html>	Tells the browser this is an HTML5 document
<html>	Root element; contains everything
<head>	Information about the page, not shown in the window: <title> (text on the tab), <meta>, <link>, <style>, <script>
<body>	The visible content: text, images, links, tables, forms

Important tags

Purpose	Tag	Notes
Headings	<h1> ... <h6>	h1 is the largest, h6 the smallest
Paragraph, line break, line	<p>, , <hr>	br and hr are empty tags
Bold, italic, underline	 or , <i> or , <u>	strong and em also carry meaning
Superscript, subscript	<sup>, <sub>	x ² , H ₂ O
Hyperlink	text	target="_blank" opens in a new tab
Image		alt = text shown if the image fails; also for screen readers
Unordered (bullet) list	with	Bullets
Ordered (numbered) list	with	Numbers; type and start attributes
Table	<table>, <tr> (row), <th> (heading cell), <td> (data cell)	colspan and rowspan merge cells
Form and inputs	<form>, <input>, <textarea>, <select>, <button>	See below

Purpose	Tag	Notes
Grouping	<div> (block), (inline)	Used with CSS
Semantic HTML5	<header>, <nav>, <main>, <section>, <article>, <aside>, <footer>	Give meaning to parts of a page
Media	<audio>, <video>, <iframe>	Play sound and video, embed another page

Table and form

```

<table border="1">
  <tr><th>Name</th><th>Marks</th></tr>
  <tr><td>Ravi</td><td>85</td></tr>
</table>

<form action="login.php" method="post">
  Name: <input type="text" name="user">
  Password: <input type="password" name="pw">
  <input type="checkbox" name="c1"> Remember me
  <input type="submit" value="Send">
</form>

```

Forms: **action** = the page that processes the data; **method** = **GET** (data in the URL, visible, for searches) or **POST** (data in the request body, for passwords and big data). Common input types: text, password, radio (choose one), checkbox (choose many), submit, reset, file.

CSS: Cascading Style Sheets

CSS controls the **look** of HTML: colours, fonts, spacing, layout. HTML = structure, CSS = style, JavaScript = behaviour. A rule has a **selector** and **declarations**: **p { color: blue; font-size: 14px; }**

Way of using CSS	Where	Example
Inline	In the tag's style attribute	<p style="color:red">
Internal	In a <style> block in the <head>	<style> p {color:red} </style>
External	In a separate .css file linked from the page	<link rel="stylesheet" href="style.css">

BPSC POINT

Priority when rules clash: **inline > internal > external** (the closest to the element wins). One external file can style a whole website, which is the best practice.

8.8 XML: eXtensible Markup Language**● LEVEL 2 · EXAM**

XML (W3C, 1998) is a markup language to **store and transport data**. Its tags are **not predefined**: you invent your own, so the data **describes itself**. XML does not show anything by itself; another program reads it.

XML example

```

<?xml version="1.0" encoding="UTF-8"?>
<students>
  <student id="1">
    <name>Ravi</name>
    <marks>85</marks>
  </student>
  <student id="2">
    <name>Sita</name>
    <marks>91</marks>
  </student>
</students>

```

Rule for a well-formed XML document	Example of a mistake
Exactly one root element	Two top-level elements side by side
Every start tag has an end tag	<name>Ravi (no closing tag)
Tags are properly nested	<a>text
Tags are case-sensitive	<Name>Ravi</name>
Attribute values are in quotes	<student id=1>

A **well-formed** document follows these rules. A **valid** document is well-formed **and** also follows a **DTD** or **XML Schema (XSD)** that defines allowed tags.

Point	HTML	XML
Purpose	Display data and decide its look	Describe, store and carry data
Tags	Predefined (p, h1, a ...)	User-defined
Strictness	Forgiving; browsers fix small errors	Strict; one error stops the parser
Case	Not case-sensitive	Case-sensitive
Closing tags	Some empty tags need none (br)	Every tag must be closed
Output	Visual web page	No output; used by programs

Uses of XML: exchanging data between different systems and platforms, configuration files, RSS feeds, Office file formats (docx, xlsx), SVG images and **web services (SOAP)**. **JSON** is a lighter alternative widely used today.

8.9 Web services and e-commerce on the web

● LEVEL 2 · EXAM

A **web service** is a program on a server that other programs call over the network, usually with **HTTP** and data in **XML or JSON**. It lets different applications, written in different languages on different computers, work together, for example a travel site asking an airline system for ticket prices, or a payment gateway confirming a payment.

Style	Idea
SOAP	Messages in XML ; strict rules; described by WSDL
REST	Uses ordinary HTTP methods (GET, POST, PUT, DELETE) on URLs; data often in JSON ; light and popular
API	Application Programming Interface: the set of rules for calling a service

E-commerce type	Between	Example
B2C	Business and consumer	Online shopping app
B2B	Business and business	Wholesale supplier portal
C2C	Consumer and consumer	Second-hand marketplace
G2C	Government and citizen	Online certificates, tax filing, bill payment

Online payment methods in India include **UPI**, debit and credit cards, net banking, wallets and cash on delivery. Secure sites use **HTTPS** and a payment gateway (security details are in the next chapter).

8.10 Common mistakes

● LEVEL 2 · EXAM

COMMON MISTAKE

- Saying Internet and WWW are the same. WWW is a **service** on the Internet.
- Saying HTML is a programming language. It is a **markup** language.
- Putting visible text in <head>. Only the <title> (shown on the tab) is meant for the head; page content goes in <body>.
- Forgetting the closing tag or nesting tags wrongly (<i>text</i>).
- Confusing domain name, URL and web hosting.
- Saying a search engine and a browser are the same thing.
- Writing an XML document with two root elements or unclosed tags: it is not well-formed.
- Thinking incognito mode hides you from websites and the ISP. It only skips saving history locally.

Quick-Solve Tricks — Summary

Situation	Trick
WWW	Tim Berners-Lee, CERN, 1989-91; a service on the Internet
Web needs	Browser + web server + HTTP + HTML + URL
URL parts	scheme://host:port/path?query#fragment; http 80, https 443
Static vs dynamic	Static = fixed HTML; dynamic = server script + database
Domain	TLD .com/.org/.net; .in country code; .gov.in; letters, digits, hyphen only
Browser vs server	Chrome/Firefox vs Apache/Nginx/IIS
Search engine	Crawler → index → ranking (PageRank)
Hosting	Shared (cheap) · VPS · Dedicated · Cloud · Colocation
HTML	Markup, tags; head (title) and body; h1-h6, p, br, hr, a, img, ul, ol, table, form
CSS	Inline > internal > external
Form method	GET shows data in URL; POST hides it in the body
XML	User-defined tags, one root, all tags closed, case-sensitive, stores data
Web services	SOAP (XML, WSDL) and REST (HTTP, JSON); API

BPSC QUICK CHECK — Chapter 8

WWW → Tim Berners-Lee	URL → address of a web resource
HTML → markup language, tags	XML → user-defined tags, stores data
<a href> → hyperlink	 → image
Static → same for all	Dynamic → database driven
Apache, Nginx, IIS → web servers	Chrome, Firefox, Edge → browsers

Traps: Internet vs WWW · domain vs URL vs hosting · HTML vs XML · head vs body · well-formed vs valid XML · browser vs search engine.

Practice questions

● LEVEL 1 · BEGINNER

- **Q1.** Expand WWW, URL, HTML, XML and HTTP.
- **Q2.** Who invented the World Wide Web, and where?
- **Q3.** What is a web browser? Name two.
- **Q4.** Which HTML tag is used to create a hyperlink?
- **Q5.** What is the first page of a website called?
- **Q6.** What is web hosting?

● LEVEL 2 · EXAM

- **Q7.** Identify the parts of `https://www.example.com:8080/docs/page.html?id=5#top`.
- **Q8.** Give three differences between a static and a dynamic web page.
- **Q9.** Give three differences between HTML and XML.
- **Q10.** Write HTML to show the image `logo.png` (alt text "Logo") that links to `https://example.com`.
- **Q11.** Name the three ways of using CSS and state which has the highest priority.
- **Q12.** Differentiate between a domain name and a URL.

● LEVEL 3 · BPSC / PGT

- **Q13.** Which of these XML fragments is well-formed? (a) `<a><b>text</a></b>` (b) `<a><b>text</b></a>` (c) `<a>text</A>` (d) `<a>1</a><b>2</b>`
- **Q14.** Write HTML for a table with a heading row (Item, Price) and one data row (Pen, 10).
- **Q15.** List the steps from typing a URL to seeing the page.
- **Q16.** Compare shared, VPS and dedicated hosting on cost, control and performance.
- **Q17.** What will this code show? `<h2>Menu</h2><ol><li>Tea</li><li>Coffee</li></ol>`
- **Q18.** Which of these are valid domain names: `my-site.in`, `my_site.com`, `-start.com`, `site123.co.in`?

Answers

Q	Solution
1	World Wide Web; Uniform Resource Locator; HyperText Markup Language; eXtensible Markup Language; HyperText Transfer Protocol.
2	Tim Berners-Lee, at CERN (1989–91).
3	Client software that requests and displays web pages, for example Chrome and Firefox.
4	The anchor tag: <code>text</code> .
5	The home page (often index.html).
6	A service that gives space on an always-connected web server so a website is visible on the Internet.
7	Scheme https ; host www.example.com (subdomain www, domain example, TLD com); port 8080 ; path /docs/page.html ; query string id=5 ; fragment top .
8	Static: fixed content, plain HTML, fast and cheap. Dynamic: created per request, uses server-side script and database, content can differ for each user.
9	HTML displays data with predefined tags and forgives errors. XML describes and carries data with user-defined tags, is strict and case-sensitive, and shows no output.
10	<code></code>
11	Inline, internal and external. Inline has the highest priority.
12	A domain name (example.com) is the name of a site. A URL is the full address of one resource: it also has the protocol, path and more (https://example.com/page.html).
13	Only (b) . (a) wrong nesting, (c) case mismatch, (d) two root elements.
14	<code><table border="1"><tr><th>Item</th><th>Price</th></tr><tr><td>Pen</td><td>10</td></tr></table></code>
15	(1) Browser asks DNS for the IP of the host. (2) TCP connection to the server (port 80 or 443; TLS first for https). (3) Browser sends an HTTP GET request. (4) Server finds the file or runs a script and returns an HTTP response with the HTML. (5) Browser reads the HTML, fetches CSS, images and scripts, and renders the page.
16	Shared: cheapest, least control, performance depends on other sites. VPS: medium cost, more control and guaranteed share. Dedicated: highest cost, full control, best performance.
17	A heading "Menu" followed by a numbered list: 1. Tea, 2. Coffee.
18	Valid: my-site.in and site123.co.in . Invalid: my_site.com (underscore not allowed) and -start.com (cannot start with a hyphen).

REMEMBER

Next: Chapter 9 — Network Security (threats and malware, hacking, phishing, firewalls, cryptography, digital signature and certificates, authentication, cyber crime and IT Act, safe browsing).

Roadmap: Ch 10 Mobile Telecom (1G to 5G), Cloud Computing and Internet of Things.